

CS4230 – Cybercrime and digital forensics

Project Assignments – Theme IV

Cybersecurity risks for organizations

Mathias LEFEBVRE
malef6335@usn.no

University of South-Eastern Norway, Campus Kongsberg

May 20, 2026

Abstract

This forensic report examines whether the user of the disk image Phishy.E01 was involved in phishing activity and identifies possible targets. One phishing-related email was recovered from the sent folder of the account sfina0919@gmail.com, addressed to Sonny Johansson and containing a spoofed link while impersonating a Skoda-related sender. A second phishing message was recovered in browser session artifacts, associated with the address mats_BRF@mail.com and addressed to Viktor Ljunggren, indicating preparation of a phishing email but not proving that it was sent. Additional findings showed the presence of concealment-related artifacts, including encryption and steganography-related traces, but these were not directly linked to the phishing activity.

Contents

1	Introduction	2
2	Evidence examined and method	2
3	Findings	2
3.1	VeraCrypt configuration	2
3.2	Disguised encrypted container	2
3.3	VeraCrypt-related documentation	3
3.4	Image-based hidden artifacts	3
3.5	Manual inspection of image data	3
3.6	Windows Mail artifacts	3
3.7	Recovered credential document	3
3.8	Mailbox access results	4
3.9	Web session artifact	4
3.10	Target information spreadsheet	4
3.11	Web history related to concealment and carding	4
4	Analysis	4
5	Conclusion	6
6	Glossary	6
7	Appendix	7

1 Introduction

A forensic examination was conducted on the disk image Phishy.E01 using the forensic software platform Autopsy. The disk image is from the computer of someone suspected of crafting phishing emails and sending those to specific victims. This report will try to find evidences and answer to these questions :

- Has the computer user been sending phishing emails ?
- Who are possible targets of those emails ?

For more information, all specific terms and tools are defined in details in the Glossary.

2 Evidence examined and method

This forensic report was prepared for an academic assignment. The findings are presented as objectively as possible and distinguish between observed facts, interpretations, and final conclusions. Digital evidence was assessed with attention to evidence integrity, consistent identification of artifacts. Any personal information encountered during the examination was considered only if it was necessary to answer the assignment questions.

The examination was done on the disk image Phishy.E01, with primary analysis performed in Autopsy. The examination focused on the main Windows installation located on volume 3 (NTFS), and in particular on the user profile Users\Serafina, where the majority of relevant artifacts were located. The analysis was performed in a structured way to identify evidence relevant to the questions. The review included file system browsing, keyword searching, inspection of browser artifacts, review of email-related storage locations, analysis of deleted and temporary files, and examination of user application data.

Additional tools were used where specific file types or artifacts required further validation, including:

- ExifTool for metadata review.
- ESEDatabaseView for Windows Mail database inspection.
- Zsteg for steganographic analysis of image files.
- John The Ripper and Hashcat for password recovery attempts.
- VeraCrypt for opening encrypted files.

3 Findings

3.1 VeraCrypt configuration

VeraCrypt was identified as installed and executed on the examined system at `/img_Phishy.E01/vol_vol3/ProgramFiles/`. A configuration file was located in the VeraCrypt application data directory inside `AppData/Roaming/VeraCrypt`. The configuration file showed that the setting *volume history logging* was disabled and the setting *automatic dismount* was configured after 60 minutes of inactivity.

3.2 Disguised encrypted container

A file named `cards.odt` was located on the file system at `/img_Phishy.E01/vol_vol3/Users/Serafina/Documents/cards.odt` and initially appeared to be an OpenDocument text file based on its extension. However, further inspection showed that the file was encrypted due to its high entropy and did not behave as a normal LibreOffice document. Instead it was identified as a VeraCrypt encrypted container. The outer container password "Artemis" was recovered during examination with Hashcat dictionary brute-force, and the mounted volume contained six images depicting rubber ducks.

3.3 VeraCrypt-related documentation

A document named `disk_encryption_v1_2.pdf` was found in VeraCrypt-related data at `/img_Phishy.E01/vol_vol3/ProgramFiles/VeraCrypt/docs/EFI-DCS` and was exported for review. The document contained technical material relating to VeraCrypt disk encryption and hidden operating system configuration. Even its presence may indicate that the user had access to documentation concerning advanced VeraCrypt features. No hidden operating system was identified on the examined disk image during this examination.

3.4 Image-based hidden artifacts

OpenStego was identified as installed on the system at `/img_Phishy.E01/vol_vol3/ProgramFiles(x86)/OpenStego`. Several image files located in `/img_Phishy.E01/vol_vol3/Users/Serafina/Pictures` were therefore examined in more detail.

A file named `hidden1.JPEG`, was found to be identical to another JPEG file in the user's Pictures folder (`/img_Phishy.E01/vol_vol3/Users/Serafina/Pictures/SavedPictures`) named `71482768873__32011EEF-C0A1-4938-A39A-64F6305B1A23.jpg`. Nevertheless the string "hidden in data" was found inside the `hidden1.JPEG` and not on the other file.

A second file named `hidden2.png` was analysed with The steganographic analysis tool. Analysis of the png file recovered the string "hiddeninthewoods". An extraction attempt using OpenStego and the recovered string as a passphrase did not produce a readable output file during this examination.

A third file, `hidden3.JPEG`, contained the metadata values "HIDDENINMETA" in the Copyright and Rights fields. The image metadata indicated capture on an Apple iPhone 15 running iOS 17.1.2 on 5 December 2023 at 14:50:42, and no GPS coordinates were present.

3.5 Manual inspection of image data

Notepad++ session have been found in `/img_Phishy.E01/vol_vol3/Users/Serafina/AppData/Roaming/Notepad++`, the data showed that the file corresponding to `hidden1.JPEG` had been opened directly in Notepad++ and that the cursor position was located far into the file contents rather than near the beginning of the file.

3.6 Windows Mail artifacts

The Windows Mail storage location `AppData/Local/Comms/UnistoreDB` was examined. The `store.vol ESE` database did not contain relevant message or recipient data during this examination, and the associated data subfolder was empty.

No direct evidence of sent phishing or spam email messages was recovered from this source. This may indicate that Windows Mail was not actively used on the system, or that related data was not present in recoverable form at the time of examination.

3.7 Recovered credential document

A password-protected OpenDocument file named `accs2.odt` was found in `vol_vol3/Users/Serafina/Downloads/`. The password hash extracted from this file was processed with John the Ripper and Hashcat, and the password "hades" was successfully recovered.

After opening the document, three sets of credentials were found inside. Two email addresses were written explicitly: "serafina_olymp@proton.me" and "sfina0919@gmail.com", each associated with a password. A third entry contained the text "bing ai", with the ProtonMail address and a different password. Further investigation showed that this email is associated to a Microsoft account.

3.8 Mailbox access results

Following recovery of the credentials from `accs2.odt`, access attempts were made to the accounts referenced in the file. Access to `serafina_olymp@proton.me` was successful, but no material judged relevant to the examination was identified in that mailbox during review.

Access to `sfina0919@gmail.com` was also successful. Within that mailbox, one email addressed to Sonny Johansson was found in the sent mail folder and was preserved as an evidence item in the appendix. In this mail the user `sfina0919@gmail.com` invites the recipient to click on a link (cf. Figure 1: 7). The provided link in the email did not redirect to any webpage and the domain name is not owned by anyone.

The access to the Microsoft account was successful. But no material were judged relevant to the examination.

3.9 Web session artifact

During investigation, a email creation and sign-up to mail provider mail.com was found. The name "Mats", the last name "BRF" and "0789685931" as cellphone number. The keyword search for "mats" return matches in browser session file, including `Session_13351859259953903` and `Session_13351859169852505` inside the `/img_Phishy.E01/vol_vol3/Users/Serafina/AppData/Local/Microsoft/Edge/UserData/Default/Sessions` folder. These files contained the address `mats_BRF@mail.com` and text corresponding to an email body was signed Mats (Chairperson) and addressed to "Viktor" inviting him to click on a link `brfvardaren4.se` for the preparation of an event (cf. figure 2: 7). The link do not redirect to any website and the domain name is not owned by anyone.

3.10 Target information spreadsheet

A spreadsheet file named `todo.ods` was found in `vol_vol3/Users/Serafina/Documents/cool/`. The file was opened during the examination and contained OSINT information concerning Sonny Johansson and Viktor Ljunggren. The contents of the spreadsheet corresponded to information already seen in the web history, including names and other target-related details (cf. figure 3: 7).

3.11 Web history related to concealment and carding

Web history entries showed searches relating to VeraCrypt, OpenStego, and carding prior to or during the period in which other relevant artifacts were created or accessed.

4 Analysis

The findings indicate that the examined computer may have been used to prepare phishing activity directed at specific individuals, commonly referred to as spear phishing. The strongest evidence is the recovery of one phishing email from the mailbox `sfina0919@gmail.com`, where the message was addressed to Sonny Johansson impersonating a Skoda service sender and encouraged the recipient to click on a spoofed Atteviks link (`www.atteviks.se`).

A second message was recovered from Edge browser session artifacts rather than from a sent mailbox. The message originated from the address `mats_BRF@mail.com`, was sent to Viktor, and was signed "Mats (Chairperson)." It also contained a link to `brfvardaren4.se`, impersonating the website of the Swedish housing cooperative BRF. The browser autofill and session data relating to the created email `mats_BRF@mail.com` support the conclusion that a phishing email with identity spoofing was created or prepared on the system. Nevertheless, because this second message was recovered from session files and not from a sent folder or recipient mailbox, the evidence supports preparation or drafting of the email, but does not by itself prove that the email was sent.

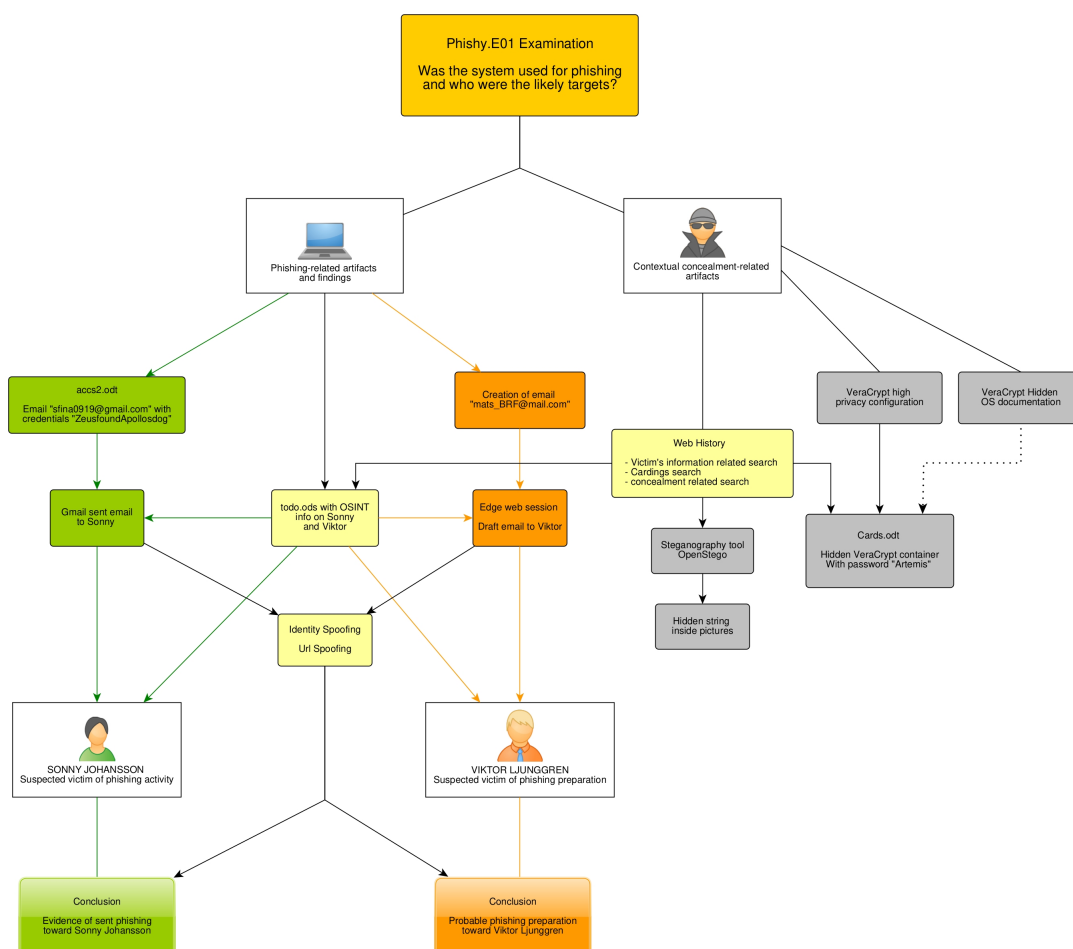
The two likely targets of this spear phishing activity are Sonny Johansson and Viktor Ljunggren. This assessment is supported by multiple artifact types: web history showing searches on both individuals, a local spreadsheet file `todo.ods` containing information about the victims, and email or draft-email content addressed directly to

them. The repetition of the same names across browser history, local documents, recovered credentials, and email-related artifacts strengthens the association between the user activity and those two persons. No direct evidence was recovered showing that credentials or data were successfully obtained from either target.

Additionally, the browser history showed searches relating to carding, VeraCrypt, and OpenStego before or around the same period in which encrypted and concealed files were identified on the system. First, the specific configuration of VeraCrypt uses settings that are not enabled by default and indicate that VeraCrypt had been configured for strong privacy use.

Also the presence of carding-related search activity is notable because the encrypted VeraCrypt container `cards.odt` was also recovered from the user's Documents folder. The content of the mounted outer volume with pictures of rubber ducks appeared benign and may represent decoy content due to its small size (1Kb) compared to the overall size of the volume (24Mb). However no hidden volume could be confirmed or excluded during examination. So the relationship between the carding-related searches and the `cards.odt` file cannot be established with certainty on these findings alone.

Finally, even if it is not directly related to phishing activity, some findings showed possible concealment activity on the system. These findings included the presence of OpenStego, unusual strings recovered from image files, metadata anomalies, and evidence that one image file had been opened in Notepad++ for raw-data inspection. However, no recovered content directly connected these artifacts to the phishing emails, the identified targets, or the suspected sending activity. For that reason, these findings are treated as contextual rather than direct evidence of phishing.



5 Conclusion

For those reasons, the forensic examination of Phishy.E01 supports a conclusion of confirmed phishing activity directed at Sonny Johansson, based on the email recovered from the Gmail sent folder. The examination also supports probable phishing preparation directed at Viktor Ljunggren, based on the draft message recovered from browser session artifacts. Sonny Johansson and Viktor Ljunggren were identified as the most likely targets of the activity.

6 Glossary

- Autopsy : The forensic software platform used to examine the disk image and review files, artifacts, and timelines.
- Artifact : A piece of digital evidence left by user or system activity, such as a file, log entry, or browser record.
- Browser session artifact : Data stored by a web browser that can preserve information about active or recent browsing sessions, such as open tabs, typed content, or session state.
- Disk image : A forensic copy of a storage device or partition created for examination without working directly on the original media.
- Entropy : A measure often used in file analysis to indicate how random or encrypted file contents appear.
- Hashcat : A password recovery tool commonly used to test or recover passwords from extracted hashes.
- Identity spoofing : The act of pretending to be another person or entity in order to deceive a target, for example by using a false sender identity in an email.
- John the Ripper : An open-source password auditing and recovery tool used to identify weak passwords or crack protected files from extracted hashes.
- Metadata : Data that describes another file, such as creation details, device information, or embedded fields.
- Steganography : A technique for hiding data inside another file, such as embedding secret content within an image.
- OpenStego : A steganography tool used to hide or extract data in image files.
- Spear phishing : A targeted form of phishing directed at a specific person or small group rather than a broad set of recipients.
- OSINT : Stand for Open-Source INTelligence, it represent all information gathered from publicly available sources such as websites, directories, and social media.
- Phishing : A form of social engineering in which a fraudulent message is used to trick a victim into revealing sensitive information or clicking a malicious or deceptive link.
- VeraCrypt : Encryption software used to create encrypted containers or volumes for protecting stored data.
- Volume : A logical storage unit or partition within a disk image.

7 Appendix

Phishing Email screenshot

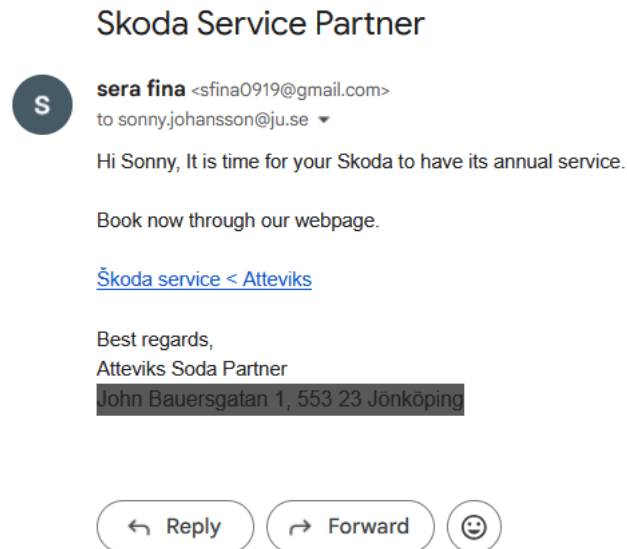


Figure 1 : Mail Send to Sonny Johansson on Gmail

Draft phishing Email body

Hi Viktor,

It is time for the annual cleaning day in BRF Vårdaren 4.

We will meet up at 9:00 at april 5th, and divide roles.

We hope to be done by 12 and then have a joint lunch.

please sign up and enter your food preferences at brfwardaren4.se

best regards,

Mats (Chairperson)

Figure 2 : Written email aimed to Viktor

[illegible]

Figure 3 : OSINT file about victims